

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: INTERNET PROGRAMMING

COURSE CODE: CSA4305

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COURSE OUTCOME COVERED

CO-2 : Apply CSS layout and styling techniques to create visually appealing and responsive web interfaces, and use JavaScript to enable interactive client-side behavior. (L3)

Assessment Tool : Short Answer Test

Weightage: 10%

QUESTIONS:

Total Marks: 20

Question 1 (5 Marks)

Suppose that a company wants its website to work seamlessly on mobile, tablet, and desktop devices. Explain the concept of responsive web design and its importance. Describe the key CSS techniques used to achieve responsiveness, and explain how CSS Flexbox or Grid can be used to design adaptive layouts. Illustrate your answer with a simple example layout and suggest improvements to enhance user experience across different devices.

Question 2 (5 Marks)

Suppose that a webpage contains a form where user input must be validated before submission. Explain the concept of client-side validation and its purpose. Describe how JavaScript event handling is used in validation and explain the role of regular expressions in validating inputs such as email or phone numbers. Provide a suitable example and suggest improvements for better usability and security.

Question 3 (5 Marks)

Suppose that a webpage layout breaks when the browser window is resized. Explain the CSS box model and its components, and describe how CSS positioning techniques affect layout structure. Identify possible causes of layout breaking and explain how media queries can be used to resolve these issues. Suggest best practices to ensure a stable and responsive layout.

Question 4 (5 Marks)

Suppose that a website requires dynamic content updates without reloading the entire page. Explain the concept of the Document Object Model (DOM) and its role in web development. Describe how JavaScript can be used to manipulate the DOM for updating webpage content dynamically. Provide an example and suggest improvements to enhance performance and user experience.

Code of Conduct:

I _P.Harika_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with	Incorrect or irrelevant answer

				noticeable errors	
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	10	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

- 1) Suppose that a company wants its website to work seamlessly on mobile, tablet and desktop devices. Explain the concept of responsive web design and its importance. Describe the key CSS techniques used to achieve responsive design, and explain how CSS Flexbox or Grid can be used to design adaptive layouts. Illustrate your answer with a simple example layout and suggest improvements to enhance user experience across different devices.

Responsive Web Design (RWD)

Responsive Web Design is a technique that makes a website automatically adjust to different screen sizes such as mobile, tablet, and desktop.

Importance:

- Improves user experience.
- Works on all devices.
- Reduces scrolling and zooming
- Better SEO and faster maintenance.

CSS Techniques

- Flexible layouts using Flexbox or Grid
- Responsive units (% , vw, vh, rem)
- Media Queries
- Flexible images (max-width: 100%;)

Example (Flexbox)

```
<!DOCTYPE html>
<html>
<head>
<style>
  .container {
    display: flex;
    flex-wrap: wrap;
  }
  .box {
    flex: 1;
    padding: 20px;
```

```
background: lightblue;
```

```
margin: 10px;
```

```
{
```

```
@media(max-width: 768px) {
```

```
  .container {
```

```
    flex-direction: column;
```

```
  }
```

```
}
```

```
<style>
```

```
</head>
```

```
<body>
```

```
<div class="container">
```

```
<div class="box">Home</div>
```

```
<div class="box">About</div>
```

```
<div class="box">Contact</div>
```

```
</div>
```

```
</body>
```

```
</html>
```

• Improvements:

- Use mobile-first design
- Optimize images
- Use readable fonts.

2.) Suppose that a webpage contains a form where user input must be validated before submission. Explain the concept of client-side validation and its purpose. Describe how JavaScript event handling is used in validation and explain the role of regular expressions in validating inputs such as email or phone numbers. Provide a suitable example and suggest improvements for better usability and security.

Client-side Validation

Client-side validation checks user input in the browser before sending data to the server.

Purpose.

- Prevent invalid input.
- Improve user experience
- Reduce server load.

JavaScript Event Handling

The submit event checks the form before submission.

Regular Expression.

Regex validates formats like email and phone numbers

Example

```
<form onsubmit="return validate()">
  <input type="email" id="email" placeholder="Email">
  <input type="submit">
</form>
```

```
<script>
```

```
function validate(){
```

```
let email = document.getElementById("email").value;
```

```
let pattern = /^[^|\s@]+@{^[|\s@]+\.{^[|\s@]+}$
```

```
if (!pattern.test(email)){
```

```
  alert("Invalid Email");
```

```
  return false;
```

```
}
return true;
```

```
}
```

```
</script>
```

Improvements

- Show error messages instantly
- Validate on both client and server.
- Use secure HTTPS.

3. Suppose that a webpage layout breaks when the browser window is resized. Explain the CSS box model and its components, and describe how CSS positioning techniques affect layout structure. Identify possible causes of layout breaking and explain how media queries can be used to resolve the issues. Suggest best practices to ensure a stable and responsive layout.

CSS Box Model

The CSS Box Model defines how elements occupy space

Components:

- Content
- Padding
- Border

CSS Positioning

- static
- relative
- absolute

Causes of Layout Breaking

- Fixed widths
- Large images
- Incorrect positioning

Media Query Example

• container {

display: flex;

{

@media (max-width: 768px) {

• container {

flex-direction: column;

}

}

Best Practices

- Use Flexbox or Grid
- Use responsive units (%), (rem).
- Apply media queries.

4. Suppose that a website requires dynamic content updates without reloading the entire page. Explain the concept of the Document Object Model (DOM) and its role in web development. Describe how JavaScript can be used to manipulate the DOM for updating webpage content dynamically. Provide an example and suggest improvements to enhance performance and user experience.

Document Object Model (DOM)

The DOM is a tree structure representing a webpage. JavaScript uses the DOM to access and modify webpage content.

Role in Web Development

- Change text
- Change styles
- Add or remove elements

Example

HTML

```
<p id = "demo"> Welcome </p>
<button onclick = "changeText()"> Click </button>
<script>
function changeText() {
document.getElementById("demo").innerHTML =
    "Welcome to Java Script!";
}
```

}

</script>

Performance & User Experience Improvements

- Update only required elements
- Minimize unnecessary DOM changes.
- Use event delegation.
- Keep JavaScript optimized.